

NITISH KUMAR

+91-9631185916 nitishkumar7791@gmail.com linkedin.com/in/Nitishkumar github.com/nitishkumar1407

Summary

An **AI/ML** and frontend developer with hands-on experience in building multiple **real-world projects** using Machine Learning and Computer Vision. Strong skills in **Python, Java**, Data Structures and Algorithms, with expertise in developing efficient, scalable, and user-friendly solutions. Currently contributing to a **patent-filed** project focused on a **multimodal AI system**.

Technical Skills

Programming Languages: Java, Python

Web Development: HTML, CSS, JavaScript

Databases: MySQL

Tools & Frameworks: Git, GitHub, Streamlit, Tkinter

Data Analysis: Pandas, NumPy, Matplotlib

Machine Learning: Random Forest, XGBoost, Linear Regression, Logistic Regression, Scikit-learn

Computer Vision: YOLOv8, FaceNet

Projects

Energy Forecasting System with Real-Time Weather & Solar API Integration | [GitHub](#)

- Developed a predictive model to forecast daily city-level energy consumption using historical temperature data and real-time **weather & solar APIs**.
- Implemented a stacked ensemble model using **Random Forest** and **XGBoost** as base models with **Linear Regression** as the meta-model to improve energy consumption forecasting accuracy, trained on a dataset of more than **30,000 records**.

Wildlife Poaching Detection System | [GitHub](#)

- A **real-time cost-effective** wildlife poaching detection system using computer vision for **automated monitoring** of protected forest areas.
- Integrated **YOLOv8** object detection and **FaceNet** face recognition models with backend processing for **AI-based image** analysis and threat detection.

Audio Language Model | [GitHub](#)

- Designed and implemented an Audio Language Model (ALM) that integrates **speech recognition**, **speaker diarization**, **audio event detection**, and **emotion analysis** for intelligent audio understanding.
- A multimodal AI pipeline integrating **Whisper** (ASR), **Pyannote**(speaker diarization), audio event and emotion models with a **fusion-based** reasoning system for contextual audio understanding.

Education

B.Tech in Artificial Intelligence & Machine Learning

2023 – 2027

Lakshmi Narain College of Technology (LNCT), Bhopal, MP

CGPA (till 5th semester): 8.16/10

Experience

AHNA Model - Patent Ongoing project

2026

- Developing AHNA, a multimodal AI system for real-time audio understanding using speech recognition and audio event detection.

Smart India Hackathon

2025

- Qualified the college-level round of Smart India Hackathon 2025 with an AI-driven solution for a real-world government challenge.

Code for Change AI Hackathon

2025

- Collaborated with a team during Code for Change AI Hackathon 2025 to develop AI-based solutions, gaining hands-on experience in rapid project development and problem-solving.